

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE:CSA1610

COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

<i>Criteria</i>	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Tejaswini.M 192411079(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Answer

The **Elbow Method** is used to determine the appropriate number of clusters in K-Means. It plots the number of clusters against the Sum of Squared Errors (SSE).

From the given values:

- $K = 1 \rightarrow SSE = 600$
- $K = 2 \rightarrow SSE = 350$
- $K = 3 \rightarrow SSE = 220$
- $K = 4 \rightarrow SSE = 200$
- $K = 5 \rightarrow SSE = 190$

The SSE decreases significantly from **K=1 to K=3**. After K=3, the reduction becomes much smaller.

Therefore, the **elbow point is K = 3**.

Patient Segmentation

The three clusters can represent different patient groups such as:

- **Cluster 1 – Low/Moderate Expense & Higher Risk:** Patients such as P1 and P4.
- **Cluster 2 – High Expense & Lower Risk:** Patients such as P2 and P5.
- **Cluster 3 – Moderate Expense & High Risk:** Patient such as P3.

Healthcare Decision-Making

Using three clusters allows the hospital to provide different strategies for different patient groups.

For example:

- High-risk patients can receive closer monitoring.
- High-expense patients can be considered for cost-management programs.
- Lower-risk patients can receive routine healthcare services.

Conclusion

The optimal number of clusters is **3**, because $K=3$ provides a substantial reduction in SSE while avoiding unnecessary additional clusters. K-Means with three clusters can help hospitals identify patient groups and make better decisions regarding treatment, monitoring, and resource allocation.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Answer

Hierarchical clustering groups similar data points into a tree-like structure called a **dendrogram**.

Initially, every order is treated as an individual cluster. The algorithm then combines the most similar orders step by step.

Based on the product category and order value:

- O1 and O3 are both **Electronics** and have similar order values.
- O2 and O5 are both **Clothing** and have similar order values.
- O4 belongs to **Furniture** and has a different category and order value.

Therefore, a reasonable interpretation is:

Cluster 1 — Electronics

O1 and O3

- O1 → ₹55,000
- O3 → ₹57,000

These orders represent high-value electronics purchases.

Cluster 2 — Clothing

O2 and O5

- O2 → ₹25,000
- O5 → ₹27,000

These represent medium-value clothing purchases.

Cluster 3 — Furniture

O4

- O4 → ₹30,000

This forms a separate group because its category differs from the other orders.

Therefore, an appropriate number of clusters is **3**.

Purchasing Behavior

The clusters help the retailer understand different purchasing patterns:

Cluster	Customer Behavior	Business Use
Electronics	High-value purchases	Premium product promotions
Clothing	Medium-value purchases	Discounts and offers

Cluster	Customer Behavior	Business Use
Furniture	Different product preference	Targeted furniture campaigns

Conclusion

Hierarchical clustering can divide orders into meaningful groups based on similarities in category and value. A **3-cluster solution** provides useful insight into purchasing behavior and helps the retailer design targeted marketing strategies.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Answer

Clustering can be used to divide mobile application users according to their behavior. The important features are:

- Daily usage time
- Login frequency
- In-app purchase amount

Users with similar activity patterns are assigned to similar groups.

Possible User Groups

Highly Active Users

U1, U3 and U5 have:

- High daily usage
- High login frequency
- Relatively lower purchase values

They can be considered highly engaged users.

Low-Activity / High-Purchase Users

U2 and U4 have:

- Lower daily usage
- Lower login frequency
- Higher purchase values

They may be less frequent users but valuable from a revenue perspective.

Soft Clustering

In **hard clustering**, a user belongs to only one cluster.

For example:

U1 → Cluster A = 100%

In **soft clustering**, a user can have membership probabilities in multiple clusters.

For example:

User Active User Revenue-Oriented User

U1	0.90	0.10
U2	0.25	0.75
U3	0.95	0.05
U4	0.15	0.85
U5	0.90	0.10

These values are **illustrative membership probabilities**, since the question does not provide actual calculated probability values.

Advantages of Soft Clustering

1. **Handles overlapping behavior**
Users may exhibit characteristics of multiple groups.
2. **Provides uncertainty information**
Membership probabilities indicate how strongly a user belongs to a cluster.
3. **Better customer segmentation**
Real customers do not always fit perfectly into one category.
4. **Improves personalization**
Different probabilities can be used to provide personalized recommendations.
5. **Useful for marketing**
Users who partially belong to multiple segments can receive targeted campaigns.

Conclusion

Soft clustering is useful for mobile applications because user behavior can overlap between different groups. Instead of forcing each user into a single category, probability-based membership provides a more flexible and realistic representation of customer behavior.

Q4. Dataset: Agricultural Crop Monitoring

Farm Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Answer

Clustering algorithms such as K-Means often use distance calculations. Therefore, the scale of the features can strongly affect the clustering result.

Here, the features have very different ranges:

- Rainfall → approximately **280–520 mm**
- Soil fertility → **3–9**
- Crop yield → **65–90 tons**

Rainfall has a much larger numerical scale than soil fertility.

Before Normalization

Without normalization, the rainfall feature can have a stronger influence on the distance calculation.

For example:

- Difference in rainfall can be hundreds of units.
- Difference in soil fertility is only a few units.

Therefore, the algorithm may mainly group farms according to rainfall rather than considering all features equally.

After Normalization

Normalization converts features into comparable scales.

A common Min-Max normalization formula is:

After normalization, rainfall, soil fertility, and crop yield are represented on a similar scale.

This allows the clustering algorithm to consider all features more fairly.

Expected Clustering

For example, farms **F1 and F4** have:

- Relatively lower rainfall
- High soil fertility
- High crop yield

Therefore, they can form a similar agricultural-performance group.

F2 and F5 have:

- Higher rainfall
- Lower soil fertility
- Lower crop yield

Therefore, they can form another group.

F3 has intermediate values and may belong to an intermediate cluster.

Benefits of Normalization

1. Prevents large-scale features from dominating.
2. Makes distance calculations more balanced.
3. Improves cluster quality.
4. Makes different agricultural variables comparable.
5. Produces more reliable agricultural analysis.

Conclusion

Normalization is important before applying clustering when variables have different scales. In this dataset, normalization prevents rainfall from dominating the clustering process and allows soil fertility and crop yield to contribute appropriately. This results in more meaningful and reliable farm segmentation.

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48
K=3 0.68
K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Answer

The **Silhouette Score** measures how well data points fit within their assigned clusters. Its value generally ranges from **-1 to +1**.

- A value close to **+1** indicates well-separated clusters.
- A value near **0** indicates overlapping clusters.
- A negative value may indicate poor clustering.

Comparison

Number of Clusters Silhouette Score Interpretation

K = 2	0.48	Moderate clustering quality
K = 3	0.68	Best clustering quality
K = 4	0.52	Moderate clustering quality

The highest score is:

for **K = 3**.

Therefore, **3 clusters are the most suitable choice**.

Employee Segmentation

The clusters could represent performance groups such as:

Cluster 1 – High Performers

Employees with high productivity, attendance, and skill scores.

For example:

- E1
- E5

Cluster 2 – Moderate Performers

Employees with moderate productivity and skill.

For example:

- E3

Cluster 3 – Lower Performers

Employees with relatively lower productivity, attendance, and skill.

For example:

- E2
- E4

The exact cluster assignment would depend on the actual clustering algorithm and distance calculations.

Benefits to the Company

Using three clusters can help the company:

1. Identify high-performing employees.
2. Identify employees who need training.
3. Recognize employees with strong attendance and productivity.
4. Develop personalized performance improvement plans.
5. Allocate training and development resources effectively.

Conclusion

The **K=3 model** is the most suitable because it has the highest silhouette score of **0.68**. This indicates that the three clusters are better separated and more internally consistent than the K=2 and K=4 models. Therefore, the company can use three employee performance groups for better evaluation and decision-making.